UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS2750

Paper Title: Probability and Statistics

Exam Period: Summer 2026

Exam Rubric

Time Allowed: 3 hours

Exam Type: Written – Restricted

Approved Calculators: Permitted

Additional Stationery

One sheet of paper (informally called a ”crib sheet”) containing handwritten notes of student’s own choice can be brought and used as an individual reference during the exam. The handwriting must be of reasonable size, e.g., similar to that used in exam writing. The crib sheet must be treated as an incentive to lessen memorization and focus instead on understanding the class materials and developing problem solving skills; solely relying on the crib sheet is likely to lead to failing the exam.

Instructions

Answer FIVE questions out of the SIX possible.

If more then FIVE questions are answered, please clearly indicate which questions you wish to be marked. Otherwise, the first FIVE questions answered in the answer booklet will be marked.

For questions requiring you to compute some quantities (e.g., probabilities or expectations) you must show your reasoning. There is no need to compute the numbers in decimal form, e.g., it is entirely appropriate to leave an answer like $\frac{324 \cdot 142 \cdot 84}{59523 \cdot 707} + \frac{\sqrt{275}}{294}$. In general, any arithmetic expression does not need to be computed as long as it is explicit; definite integrals, however, must be solved.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

The exam question paper **MUST NOT** be removed from the examination venue.

Permitted calculators: CASIO FX82, FX83 and FX85 series.

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Question 1 [20 marks]

(a) Prove the following inequality for any events A and B: [5]

$$\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1 \, .$$

(b) Generalize and prove the inequality from (a) for n events $A_1, \ A_2, \ldots, A_n$. [15]

Question 2 [20 marks]

(a) Consider a batch of 100 items out of which 5 are faulty. Four different items of the batch are inspected at random; the batch is accepted if all inspected items are working. What is the acceptance probability of the batch? [15]

(b) For the random experiment from (a) give an example of two different events which are *not* independent. [5]

Question 3 [20 marks]

(a) A fair coin is tossed until either two consecutive Heads or two consecutive Tails appear. Compute the expected value of the number of tosses. [5]

(b) A fair coin is tossed until the appearance of a Tail immediately preceded by a Head. Compute the expected value of the number of tosses. [15]

Question 4 [20 marks]

(a) Explain how the exponential distribution can be obtained from the geometric distribution using a limiting argument. [5]

(b) A personal tutor mistakenly schedules two private (individual) meetings with two tutees for the same time. The first tutee arrives on time whereas the second tutee arrives 5 minutes late. The durations of the meetings are exponentially distributed with an average of 30 minutes and are independent. Compute the expected value of the time between the arrival time of the first tutee and the departure time of the second tutee. [15]

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Question 5 [20 marks]

(a) Describe the concepts of *limiting distribution* and *stationary distribution* in the context of Discrete-Time Markov Chains. [5]

(b) A phone breaks with probability p each day; when that happens it gets replaced. Upon functioning properly for 30 consecutive days a phone also gets replaced. What is the long-term frequency of phone replacements? [12]

(c) Estimate the same long-term frequency from (b) provided that replacements for working phones happen after 1000 days. [3]

Question 6 [20 marks]

Consider a coin showing Heads with probability p.

(a) Let $n > 0$ and N be the number of observed Heads in n tosses. Find the ML estimator for p in terms of N and determine whether it is unbiased. [10]

(b) Let $k > 0$ and K be the number of tosses until k Heads are observed. Find the ML estimator for p in terms of K. [10]

End